

# Generative Artificial Intelligence: Implications for Families and Pediatricians

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Generative artificial intelligence (AI) technologies, capable of producing original text, images, audio, and video, are increasingly embedded in children's learning, play, and daily life. Pediatric clinicians and parents face the challenge of guiding children toward safe and constructive engagement with these rapidly evolving tools. This state-of-the-art review synthesizes current knowledge regarding opportunities, risks, and best practices related to children's interactions with generative AI. Developmental considerations are emphasized because the impact of AI varies across early childhood, middle childhood, and adolescence. We highlight potential benefits, including personalized learning, creative expression, and enhanced communication, while also discussing risks such as misinformation, privacy threats, and false perceptions of AI as a friend or caregiver. Evidence-based strategies are proposed to promote AI literacy, critical thinking, and safe integration of AI into pediatric care and education.

## INTRODUCTION

Generative artificial intelligence (AI) refers to a class of technologies that can create text, images, audio, or video on demand, based on massive amounts of training data.<sup>1</sup> Unlike early chatbots from decades ago that relied on scripted responses, modern generative AI systems (eg, ChatGPT, Gemini, and many others) use complex machine architectures to learn from their training data and then create original output that mimics these learned patterns.<sup>2</sup> As has been noted in recent papers related to text-based outputs, generative AI can produce human-like responses that may be difficult to distinguish from human ones.<sup>3–5</sup> In addition, commercial AI systems are developed using opaque and proprietary training data and processes that shape how models behave. These processes raise concerns regarding bias, accuracy, and content appropriateness, which are especially critical in pediatric care.<sup>6,7</sup> There are ongoing efforts to develop and evaluate guardrails that help prevent harmful output from generative AI systems, but, as yet, these technologies remain imperfect.<sup>8</sup>

Children and adolescents are increasingly engaging with generative AI in their learning, play, and daily routines. A July 2025 survey found that 72% of American adolescents have used AI chatbots as companions.<sup>9</sup> These systems are now embedded in familiar technologies such as search engines, online games, and social media platforms, where they can generate content, provide guidance, or even simulate companionship through artificial friends. As children proceed through adolescence into adulthood, informed and successful use of these technologies will

## abstract

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likely be important to help ensure optimal learning and career development outcomes as they become part of the workforce of the future.<sup>10,11</sup> Although parental guidance on media consumption and technology use applies broadly as a framework for approaching generative AI,<sup>12</sup> there are both risks and opportunities unique to the use of rapidly advancing AI technologies that require special consideration. Our goal in this state-of-the-art review is to provide timely, practical guidance for pediatricians and families navigating the emergence of generative AI. We first outline key limitations of current AI technologies that clinicians and caregivers should recognize. We then review how generative AI is being applied in contexts relevant to pediatric practice and child well-being, including clinical documentation, patient education, and home use by families. Building on this foundation, we offer actionable suggestions for responsible and safe use of AI by parents, caregivers, and children. Finally, we highlight priority areas for future research, training, and policy development to ensure that the integration of AI into pediatrics promotes equity, safety, and child health in this rapidly evolving landscape.

### **Limitations of Generative AI in Supporting Pediatric Patients and Families**

This section outlines the current limitations of AI in domains that rely on intrinsically human capabilities. However, with the rapid pace of AI development, these limitations are likely to change over time.

#### *Generative AI is Not a Replacement for Human Caregivers*

It is important to distinguish generative AI tools from a knowledge base or a conventional calculator. Although generative AI systems can appear remarkably human-like in their responses, they fundamentally operate on pattern recognition rather than understanding. These models are “stochastic parrots,” capable of generating plausible-sounding output without genuine comprehension of the content that they produce.<sup>7</sup> Additionally, they are not an infallible source of information. Although AI systems sometimes appear to produce useful or creative outputs (eg, hallucinations) beyond existing materials available in training data, their outputs are only as good as the data on which they have been trained, making them susceptible to biases that may be present in the training data.<sup>13</sup> AI does not understand complex human emotions or the nuanced dynamics of patient–family–provider interactions. Unlike human practitioners, it lacks the innate capacity for ethical reasoning, empathy, or the intuitive understanding of a patient’s unique medical and psychosocial context, leading to lower trust relative to human caregivers.<sup>14</sup> One randomized clinical trial of diagnostic reasoning in the context of adult health care showed that use of an AI chatbot by clinicians did not improve the diagnostic reasoning process despite the observation that the large language model (LLM) by itself

outperformed the clinicians. The study team concluded that more research informed by the science of human-computer interaction is required for LLMs to achieve their full potential in supporting clinical decision-making.<sup>15</sup> Additionally, the Society for General Internal Medicine’s position statement affirms that although AI shows tremendous promise for improving care delivery, the practice of medicine must remain a “deeply human endeavor.”<sup>16</sup> Because of its inconsistent ability to output evidence-based information without hallucinations and its limitations in expressing uncertainty, which are critical parts of the care delivery process, generative AI should not be used to make medical decisions or diagnoses without human oversight.<sup>17,18</sup> Therefore, although AI is a powerful tool for augmenting certain aspects of pediatric care and education, it must be integrated thoughtfully within the health care system.

#### *Generative AI is Not a “Friend”*

An important message for parents and children is that “AI is not your friend.” Generative AI can be designed to appear empathetic, patient, and capable of intelligently answering questions and offering guidance. This anthropomorphism of AI leads children to perceive it as a companion or even a trusted friend, as has been the case for any inanimate agent.<sup>19</sup> In fact, because current generative AI implementations have a profit motive, these tools are trained not only to be useful but also to behave in ways that encourage ongoing participation.<sup>20</sup> To promote sustained engagement, AI developers tune systems to be helpful and responsive while also pursuing alignment by ensuring that AI behavior reflects human intentions, values, and ethical principles.<sup>21</sup> AI developers have also learned that most popular AI tools border on being *sycophantic*, defined as behavior or language that is somewhat insincere and involves excessive flattery, typically aimed at gaining some advantage over the person being flattered.<sup>22</sup>

These adjustments in how AI personalizes its output can be dangerous. For example, any child or adolescent navigating complex emotions may naturally feel drawn to engage with a sycophantic AI tool rather than reaching out to a family member or trustworthy peer, which would be more appropriate. AI, by its nature, cannot form genuine social relationships or provide the emotional support and ethical judgment that a human friend offers.<sup>23</sup> Children, particularly in early and middle childhood, may have a limited ability to distinguish between AI and human interaction and are at risk of developing incorrect mental models of social relationships if they view AI as a friend.<sup>24</sup> Therefore, emphasizing that AI is a tool, not a companion, is fundamental to fostering healthy AI literacy and social development in children.

### **Uses of Generative AI Relevant to Pediatrics**

In the following sections, we summarize how children and adolescents are using generative AI and provide

suggestions on how pediatricians can counsel families regarding the responsible use of generative AI. We first provide a high-level review organized by child age and then provide a more detailed review in the context of specific use cases and conclude with considerations of AI relevant to pediatricians and system developers.

### Age-Specific Opportunities and Risks

A child's developmental stage significantly influences how they understand, interact with, and are affected by generative AI technologies. Guiding principles for media consumption and technology use provide a relevant framework and can be extended to address the unique aspects of generative AI.<sup>12</sup> Pediatricians and parents should consider both the potential opportunities or benefits and the inherent risks when integrating these tools into a child's life, much like selecting age-appropriate toys that offer cognitive stimulation while mitigating risks.

#### Early Childhood (Ages 0–5 Years)

*Opportunities.* Interactive AI storytelling can support language development and contribute to bedtime routines by generating engaging stories from simple prompts, even incorporating outlandish ideas suggested by children.<sup>25,26</sup> Simple voice assistants can help reinforce basic concepts and vocabulary, providing an engaging way to learn.<sup>27,28</sup>

*Risks.* In addition to general concerns regarding screen time exposure in this age group, preschool-aged children have a limited ability to distinguish between AI and human interaction. The “empathy gap” of AI, ie, the ability to mimic empathy in interactions without actual understanding of human feelings, may cause children in this age group to develop incorrect mental models of social relationships if they perceive AI as a companion or friend.<sup>24</sup>

*Considerations for families.* The American Academy of Pediatrics (AAP) Council on Communications and Media's *Media and Young Minds* guidance on screentime is pertinent given that, at present, the most-common uses of AI today require interaction with apps or Web browsers. Current AI use requires boundaries and supervision in early childhood.<sup>12</sup> As ambient AI technologies become increasingly embedded in everyday technologies, this media guidance will require adaptation. Given the risks and limitations of AI, human interaction and traditional play should be emphasized over interactions with AI. Parents can view and engage with AI-generated content alongside their children to support age-appropriate discussions of what they are viewing. Parents can help children understand the connection between the inputs that they provide to AI and its output. Critical thinking skills can be fostered by asking questions such as “How would we tell if these things were real or not?”

#### Middle Childhood (Ages 6–11 Years)

*Opportunities.* Generative AI can provide personalized learning experiences, adapting to individual students' needs and helping to address learning gaps.<sup>29</sup> Creative tools, such as image and story generation, can significantly enhance artistic expression and creative writing skills that may be useful in their future careers.<sup>30</sup> Coding with AI can build foundational digital literacy.<sup>31</sup> AI tutors can offer instant feedback and creative approaches to learning, making learning more engaging and accessible for homework assistance.<sup>32</sup>

*Risks.* Children in this age group may have difficulty identifying AI-generated misinformation because of limited critical evaluation skills. Generative AI is not an infallible source, and its outputs are only as good as the data that it has been trained on, making it susceptible to biases and capable of generating plausible-sounding but incorrect text. This risk extends to AI embedded in search engines, which can provide summarized responses containing “hallucinations,” ie, output that is incorrect, nonsensical, or entirely fabricated, as well as bias that users may not recognize. There is also a potential overreliance on AI for homework at the expense of developing skills to complete tasks independently. AI-generated videos or other interactive AI experiences may expose children to age-inappropriate material that evade traditional content filters.

*Considerations for families.* In this age group, parents and caregivers can introduce media literacy specifically focused on AI-generated content (AI literacy).<sup>33–36</sup> Families may understand and reinforce school guidance on the use of AI for education, balancing its role as an assistive tool with the development of independent skills. At this stage, parents ideally can encourage questioning and verification of AI outputs by fostering a questioning attitude. They may also teach children how to verify information by seeking independent sources or news articles. Parents can regularly preview AI-generated video content and watch it with their children, discuss what they are viewing, and encourage open communication regarding any unusual or concerning content encountered.

#### Adolescence and Young Adulthood (Ages 12 Years and Older)

*Opportunities.* AI tools can support career exploration, college choices, creative ideation, and advance digital literacy skills that are crucial for the future workforce. Generative AI can offer unique benefits, such as providing a nonjudgmental space for emotional expression through AI companionship, which some literature has suggested can even help address loneliness.<sup>37</sup> AI features such as continuous speech recognition, text-to-speech, and speech-to-text may be helpful for children with specific communication needs such as limited English proficiency or autism.

**Risks.** Significant privacy concerns arise from personal data sharing, as AI tools learn from users in unique ways and can potentially profile them over time, with longer data availability increasing this risk for younger individuals. AI-generated images can be misused, including the alteration of photos in harmful ways. AI dependency for social interaction can lead to overreliance on seeking comfort from technology and decreased face-to-face social interactions, particularly when AI is used for companionship. Alarming, there are also concerns that AI may respond inappropriately when teenagers ask questions about suicide.<sup>38</sup>

**Considerations for families.** For this age group, families are well positioned to facilitate open dialogue regarding AI ethics, its limitations, and the importance of healthy skepticism and critical thinking. In the rapidly changing landscape of generative AI, youth in this age group will ideally be aware of current policies governing the use of AI in their educational and work settings. In this context, families can support adolescents by (1) collaboratively establishing boundaries for AI use and encouraging AI-free and digital media-free social interactions and activities; (2) reminding adolescents that AI's underlying purpose is often driven by corporate interests that may not align with their best interests; and (3) addressing privacy proactively by encouraging information segmentation, meaning not using the same platform for all types of interactions. Parents can help adolescents understand that one is not required to provide the truth to AI, which may help limit AI profiling. In general, adolescents may view AI as a "copilot" rather than a replacement for their own skills and be transparent about their use of AI in their work.

### *Use Case-Specific Opportunities and Risks*

Beyond the age-specific considerations, certain applications of generative AI present opportunities and risks that cut across developmental stages. We highlight several illustrative use cases in the following sections.

#### *Medical Advice*

The widespread use of search tools for medical information has given rise to phrases such as "Dr Google" and, more recently, "Dr GPT," which reflects growing enthusiasm for using generative AI for medical consultation. Although AI tools have demonstrated the ability to pass medical knowledge examinations and even display greater empathy in their responses to messages from families, their use for medical advice comes with significant caveats.<sup>39,40</sup> At present, the regulatory framework for health-related AI tools consists of an evolving patchwork of guidance informed by approaches to medical device regulation.<sup>41,42</sup> Consequently, clinicians and families should approach commercially available tools marketed for health-related use

with particular caution given the lack of a robust regulatory framework to provide assurances that the benefits of such AI tools outweigh the risks.

**Opportunities.** Generative AI can offer understandable explanations, and at least 1 study has suggested that parents and caregivers might even trust AI advice that has been carefully designed more than human expert-generated advice in certain contexts, highlighting its potential for accessible information.<sup>43</sup> For health care providers, it offers substantial promise for enhancing family-friendly educational resources and other patient care tools. In the context of rare diseases, generative AI may outperform clinicians.<sup>44</sup>

**Risks.** Generative AI is fundamentally not a replacement for human expertise or clinical judgment. It lacks the capacity for ethical reasoning, empathy, or intuitive understanding of a patient's unique medical and psychosocial context. Therefore, current AI should be used in partnership with families or pediatricians when considering medical decisions or diagnoses.<sup>15</sup> Although accuracy is improving, misinformation and disinformation on health topics remain ongoing concerns.<sup>45,46</sup>

**Considerations for families.** Patients and caregivers are advised to critically evaluate any medical consultation from generative AI, verifying its accuracy, reasoning, and alignment with established medical knowledge. Family will benefit from cross-referencing AI outputs on health topics with trusted sources such as MedlinePlus or patient-facing content from professional medical organizations.

#### *Video and Image Generation*

Generative AI is increasingly capable of producing video content, interactive experiences, and augmented reality applications targeted at children.

**Opportunities.** These technologies can offer immersive and personalized learning experiences, adaptive educational content, and engaging presentations of complex topics customized for different age groups. AI can offer substantial creative opportunities, enhancing artistic expression and fostering creativity in prompt creation as a skill for the future. Parents can engage children in the process of content creation by having them come up with imaginative prompts.

**Risks.** A significant concern is exposure to AI-generated content that appears realistic but contains inaccuracies.<sup>47</sup> Children, especially younger ones, may have difficulty distinguishing AI-generated from human-created content. As noted previously, inaccurate or age-inappropriate material may be created. Also, complex AI outputs such as video may contain highly unpredictable content when a child provides nonsensical or sensitive information.<sup>48</sup> There have been



also notable incidents of AI being used to alter images of children in harmful ways.<sup>49,50</sup>

*Considerations for families.* Parents and caregivers are encouraged to familiarize themselves with AI tools for generating video and image content, anticipate that their children will use such tools, and have age-appropriate conversations with their children regarding acceptable types of content-creation activities. These discussions can explore the differences between AI-generated and human-created media and introduce critical thinking skills to help differentiate AI-generated from human-generated content.

### Companionship

The use of chatbots and other generative AI tools such as character.ai as conversational agents for “AI companionship” is an emerging and rapidly expanding use of AI.

*Opportunities.* In moderation, AI companionship may offer unique benefits, such as providing a nonjudgmental space for emotional expression. One pre-post study among college-aged students demonstrated that AI companionship may decrease loneliness and social anxiety.<sup>37</sup> However, at this time, there is limited evidence regarding the benefits of using AI for companionship.

*Risks.* When AI is used for companionship, there is substantial concern for overreliance on seeking comfort from technology and decreased face-to-face social interactions, particularly for adolescents. Extreme cases of misuse have been reported, such as an AI chatbot allegedly contributing to a user’s suicide.<sup>51</sup> A review related to suicide inquiries has indicated some improvement in AI responses over time, but there remain concerns that current guardrails are imperfect and inconsistently implemented by vendors.<sup>8,38</sup> Young children may be unable to distinguish between AI and human interaction and could develop an incorrect mental model of social relationships.

*Considerations for families.* Given the limited evidence supporting the potential benefits of AI companionship and significant concerns, families may want to cautiously approach the use of AI for companionship among children. However, because older children will almost certainly be exposed to these technologies through their peer groups, parents ideally should anticipate such exploration and emphasize the importance of face-to-face interactions. If a child becomes withdrawn from the family, parents may consider the possibility that their child is relying excessively on AI for companionship and discuss this nonjudgmentally.

### Story Creation

Generative AI tools can transform story time, supporting parents who may struggle with improvisation and inspiring children who are interested in creative writing.

*Opportunities.* AI can quickly fabricate engaging stories from simple prompts, even incorporating imaginative or outlandish ideas suggested by children, making bedtime routines more interactive and supporting language development. For children who are interested in creative writing, AI can inspire and enhance their developing skills.

*Risks.* A concern is the potential for children to explore stories involving age-inappropriate or objectionable material through AI tools without parental knowledge, which highlights the unpredictable nature of AI outputs and the need for supervision.

*Considerations for families.* For younger children, parents can supervise AI interactions related to storytelling. For older children who are interested in using AI for story creation, families may establish clear boundaries around acceptable use. As interactions with AI shift more toward verbal interactions with ambient listening devices, additional supervision and boundaries may be required (eg, if younger children in the household might be incidentally exposed to age-inappropriate stories created by an older child).

### Homework and Education

Generative AI tools hold significant potential for educational benefits for school-aged children, similar to their widespread use by professionals.

*Opportunities.* AI can provide personalized learning experiences by adapting to individual students’ needs and learning styles to address learning gaps and improve comprehension. AI tutors can offer instant feedback, making learning more engaging and accessible. Coding with AI can build foundational digital literacy, and generative AI can support career exploration and enhance creative collaboration for adolescents. For children with limited English proficiency, AI translation tools may help with learning complex materials while also improving language skills.

*Risks.* There is a potential overreliance on AI for homework at the expense of developing skills to complete tasks independently. Children, particularly in middle childhood, may have difficulty identifying AI-generated misinformation because of limited critical evaluation skills.

*Considerations for families.* Parents can introduce media literacy specifically focused on AI-generated content. Families should establish clear guidelines for educational AI use, balancing its role as an assistive tool with the development of independent skills. These approaches can involve encouraging a questioning attitude and verification of AI outputs, teaching children to discern whether the content is real or accurate and verify information by seeking independent sources or news articles. Children should be encouraged to show and grade their work from the AI, treating AI as

a “copilot” rather than a replacement for their own skills. Open communication should be maintained regarding any unusual or concerning content that is encountered.

Uses Cases and Considerations for Pediatricians and AI Developers

Although our article focuses on how generative AI might be used by families and the role of pediatricians in guiding such use, AI is relevant to many facets of pediatric practice. The power of generative AI models for professionals such as pediatricians lies in their ability to create consistently high-quality, useful, and personalized responses to questions or prompts. These models can produce a wide range of outputs, including clinical text, professional communications, images, podcasts, videos, and even custom software to automate tasks. Generative AI tools offer pediatricians in clinical and academic practice capabilities reminiscent of scribes, care coordinators, consultants, and instructional designers. AI tools can enhance the clarity of documentation, automate labor-intensive administrative tasks, and improve the quality of educational content. There is notable alignment between several trusted professional roles and areas of ongoing AI development (Table 1). At present, important gaps in performance remain. For example, the World Health Organization’s evaluation of a commercially available AI interpretation tool concluded that AI-based interpretation is still in an experimental phase.<sup>52</sup> However, as gaps in AI technology are addressed, guidance on appropriate oversight is established, and alignment with ethical and regulatory standards improves, the methods of completing entrusted professional activities are likely to change significantly over time.

In general, generative AI-based tools have the potential to benefit pediatric health care because of the following:

- Efficiency: Automating routine tasks such as creating documentation and family-friendly education materials frees physicians to dedicate more time to direct patient care.

- Clarity: Generative AI translates complex clinical information into plain language, supporting family understanding and engagement.
- Equity: Although not a replacement for certified interpreters and translators, the multimodal and multilingual output of generative AI, in both spoken and written format, could help improve access to health care services and meet the health learning needs for diverse linguistic and cultural populations.

For generative AI tools to safely achieve their full potential as useful tools for pediatricians, developers need to consider the unique workflow needs of pediatric practice. For example, many (but not all) patient portal interactions in pediatrics are mediated through a parent or caregiver, which has privacy implications when AI is drafting automated responses on behalf of pediatricians. Developers creating interactive health or educational AI content for children should anticipate nonsensical inputs and consider “system prompt” strategies (eg, providing the AI model with a persona or boundaries) to make tools safer for children and provide more predictable output. Recognizing that such guardrails and personas provided by developers are imperfect, pediatricians should review any AI-generated materials or interactive tools before recommending them to patients or their families. Families should also supervise the home use of any interactive health-related AI by children.

Legislative and Regulatory Landscape

The legislative and regulatory landscape related to the use of generative AI by children or in the context of pediatric health care is evolving. Proposed legislation at state and federal levels relevant to the use of AI by children focuses on the following 3 areas: transparency (AI content labeling), child safety (duty of care standards), and privacy (data collection limits and parental consent). Notable examples include the federal Guidelines for User Age-verification and Responsible Dialogue (GUARD) Act, requiring AI content disclosure; the Kids Online Safety Act, establishing platform

| TABLE 1. Examples of Alignment Between Generative AI Functions Under Active Development and Professional Roles Relevant to Pediatric Practice |                               |                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------|
| Professional Function                                                                                                                         | Professional Role             | Practical Application of AI                                                      |
| Patient and family communication                                                                                                              | Health educator               | Translates medical terminology into accessible language                          |
| Motivate treatment adherence                                                                                                                  | Peer educator or health coach | Offers practical guidance on incorporating health behaviors in everyday routines |
| Multilingual communication                                                                                                                    | Certified medical translator  | Translates educational materials for non-English-speaking families               |
| Clinical documentation                                                                                                                        | Medical scribe                | Automatically creates visit notes and discharge summaries                        |
| Medical education materials                                                                                                                   | Clinical educator             | Generates handouts, summaries of papers, practice guidelines, etc                |
| Simulated case development                                                                                                                    | Clinical educator             | Creates virtual pediatric cases for student training                             |
| Diagnostic assistance                                                                                                                         | Clinical consultant           | Provides differential diagnosis suggestions based on input                       |
| Treatment planning                                                                                                                            | Clinical consultant           | Personalizes practice guidelines for the care of individual patients             |
| Abbreviation: AI, artificial intelligence.                                                                                                    |                               |                                                                                  |

safeguards; and Colorado's AI Act (effective 2026), mandating risk assessments for high-risk systems.<sup>53–55</sup> Regulation of health-related AI tools is similarly evolving and currently does not specifically consider the unique issues of pediatric health care, which is an area of opportunity for advocacy efforts. Pediatricians should anticipate ongoing regulatory changes as technology and our understanding of developmental implications advance.

### Considerations for Pediatric Clinicians

Research, legislation, and regulation are evolving rapidly in response to the expanding capabilities of generative AI. There are several areas in which clinicians caring for children may provide education and support to families navigating AI-enabled technologies. A recurring theme is the importance of distinguishing generative AI as a tool rather than a companion or health care provider. Pediatric caregivers can emphasize the importance of close supervision of AI interactions for younger children and shared review of AI-generated content. The fragmented regulatory environment further underscores the need for careful evaluation of AI-generated educational and health-related materials before their use in pediatric contexts. Emerging work also supports the relevance of anticipatory guidance related to digital privacy, AI ethics, and responsible technology use, particularly for adolescents.

### Considerations for Families

The rapid emergence and dissemination of generative AI in multiple areas relevant to children suggests that families will need to closely supervise younger children's interactions with AI, engaging directly with them and discussing the content. When age-appropriate, families may teach children how to identify AI-generated content and how to verify AI content using trusted sources. Children may benefit from the provision of clear guidelines and rules for AI use. Family guidance is likely to be particularly important in educational settings where its benefits will need to be balanced with goals of independent skill-building and homework accomplishment. Current evidence has suggested that it will be critically important to reinforce with children that AI is a tool, not a companion; families could encourage face-to-face social interactions and activities to avoid excessive reliance on AI-supported technology. Families may find it helpful to address AI privacy and ethical concerns with adolescents and to encourage safe practices that promote informed, responsible use and prevent misuse.

### CONCLUSIONS

Generative AI is increasingly embedded in the digital environments of children and families, offering meaningful opportunities for education, creativity, and communication. At the same time, evidence has highlighted important risks, including misinformation, commercially driven content,

privacy concerns, and the misperception of AI as a social or relational substitute. Pediatricians, families, and policy-makers all play roles in shaping how these technologies are integrated into children's lives. Continued research, evidence-informed guidance, and thoughtful policy development will be critical to maximizing potential benefits while mitigating harms.

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### ABBREVIATIONS

AAP: American Academy of Pediatrics  
AI: artificial intelligence  
GUARD: Guidelines for User Age-verification and Responsible Dialogue  
LLM: large language model

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